Lean Support for the Full Unconditional Closure Theorem

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Purpose

This note records the Lean framework in the repository that supports the "full unconditional closure" result stated in `RS_CostAlgebra_Recognition_Category.tex`, and it gives a complete written proof in ordinary mathematical language.

Supporting Lean Modules

The relevant public modules are:

- `RecognitionScience/Cost/FunctionalEquation.lean`
- `RecognitionScience/Foundation/DAlembert/Unconditional.lean`
- `RecognitionScience/Foundation/DAlembert/FullUnconditional.lean`
- `RecognitionScience/Cost/AczelTheorem.lean`

The key theorem names are:

- `dAlembert_cosh_solution_aczel`
- `washburn_zlatanovic_uniqueness_aczel`
- `J_computes_P`
- `J_surjective_nonneg`
- `P_determined_nonneg`
- `consistency_forces_RCL_form_is_theorem`
- `full_unconditional_inevitability`
- `washburn_zlatanovic_full_unconditional`

Lean Imports

```
import RecognitionScience.Cost.FunctionalEquation
import RecognitionScience.Foundation.DAlembert.Unconditional
import RecognitionScience.Foundation.DAlembert.FullUnconditional
```

Main Lean Theorem

The public theorem in the repository closest to the paper statement is `washburn_zlatanovic_full_unconditional`. For portability in this document, the statement below is written in ASCII Lean style while preserving the theorem name and hypotheses.

```
theorem washburn_zlatanovic_full_unconditional
    (F : Real -> Real)
    (P : Real -> Real -> Real)
    (hSymm : forall x : Real, 0 < x -> F x = F (1 / x))
    (hUnit : F 1 = 0)
    (hSmooth : ContDiff Real 2 F)
    (hCalib : deriv (deriv (G F)) 0 = 1)
    (hCons : forall x y : Real, 0 < x -> 0 < y ->
      F (x * y) + F (x / y) = P (F x) (F y))
    (hPoly : exists a b c d e f : Real,
      forall u v, P u v = a + b*u + c*v + d*u*v + e*u^2 + f*v^2)
    (hSymP : forall u v, P u v = P v u)
    (hNonTriv : exists x : Real, 0 < x /\ F x /= 0)
    (hCont : ContinuousOn F (Set.Ioi 0))
    (hP11 : P 1 1 = 6) :
    (forall x : Real, 0 < x -> F x = Cost.Jcost x) /\
    (forall u v : Real, 0 <= u -> 0 <= v ->
      P u v = 2 * u * v + 2 * u + 2 * v)
```

Its body is the short wrapper:

```
refine full_inevitability_explicit F P hSymm hUnit hSmooth hCalib hCons ?_ ?_
  exact consistency_forces_RCL_form_is_theorem F P hSymm hUnit hSmooth
    hCons hPoly hSymP hNonTriv hCont hP11
  exact dAlembert_forces_cosh_is_theorem
```

Auxiliary Lean Results

The two structural ingredients used by the wrapper above are:

```
theorem consistency_forces_RCL_form_is_theorem
    (F : Real -> Real) (P : Real -> Real -> Real)
    (_hSymm : forall x : Real, 0 < x -> F x = F (1 / x))
    (hUnit : F 1 = 0)
    (_hSmooth : ContDiff Real 2 F)
    (hCons : forall x y : Real, 0 < x -> 0 < y ->
      F (x * y) + F (x / y) = P (F x) (F y))
    (hPoly : exists a b c d e f : Real,
      forall u v, P u v = a + b*u + c*v + d*u*v + e*u^2 + f*v^2)
    (hSymP : forall u v, P u v = P v u)
    (hNonTriv : exists x : Real, 0 < x /\ F x /= 0)
    (hCont : ContinuousOn F (Set.Ioi 0))
    (hP11 : P 1 1 = 6) :
    forall x y : Real, 0 < x -> 0 < y ->
      P (F x) (F y) = 2 * F x * F y + 2 * F x + 2 * F y
```

```
theorem dAlembert_cosh_solution_aczel
    (H : Real -> Real)
    (h_one : H 0 = 1)
    (h_cont : Continuous H)
    (h_dAlembert : forall t u, H (t + u) + H (t - u) = 2 * H t * H u)
    (h_d2_zero : deriv (deriv H) 0 = 1) :
    forall t, H t = Real.cosh t
```

The unconditional determination of the combiner once the canonical cost is known is handled in:

```
theorem J_computes_P :
    forall x y : Real, 0 < x -> 0 < y ->
      Cost.Jcost (x * y) + Cost.Jcost (x / y) =
      2 * Cost.Jcost x * Cost.Jcost y +
      2 * Cost.Jcost x + 2 * Cost.Jcost y

theorem J_surjective_nonneg :
    forall v : Real, 0 <= v ->
      exists x : Real, 0 < x /\ Cost.Jcost x = v

theorem P_determined_nonneg (P : Real -> Real -> Real)
    (hCons : forall x y : Real, 0 < x -> 0 < y ->
      Cost.Jcost (x * y) + Cost.Jcost (x / y) =
      P (Cost.Jcost x) (Cost.Jcost y)) :
    forall u v : Real, 0 <= u -> 0 <= v ->
      P u v = 2*u*v + 2*u + 2*v
```

Mathematical Statement

Theorem 1 (Full unconditional closure)**.** *Let $F : \mathbb{R}_{>0} \to \mathbb{R}$ be C^2, and let $P : \mathbb{R}^2 \to \mathbb{R}$ be symmetric. Assume:*

(i) $F(1) = 0$,
(ii) $F(x) = F(x^{-1})$ *for all* $x > 0$,
(iii) $\left.\dfrac{d^2}{dt^2}\right|_{t=0} F(e^t) = 1$,
(iv) $F(xy) + F(x/y) = P(F(x), F(y))$ *for all* $x, y > 0$,
(v) F *is nontrivial:* $F(x_0) \neq 0$ *for some* $x_0 > 0$,
(vi) $P(1, 1) = 6$.

If, in addition, P lies in the quadratic symmetric ansatz used by the public Lean theorem,

$$P(u, v) = a + b\,u + c\,v + d\,uv + e\,u^2 + f\,v^2,$$

then

$$F(x) = \frac{1}{2}(x + x^{-1}) - 1 \qquad (x > 0),$$

and

$$P(u, v) = 2uv + 2u + 2v \qquad (u, v \geq 0).$$

Remark 1. *The public Lean development visible in the repository proves the theorem above. The d'Alembert-to-cosh step is packaged through* `aczel_representation_axiom` *in* `RecognitionScience/Cost/AczelTheorem.lean`*, and the combiner-forcing step in* `consistency_forces_RCL_form_is_theorem` *uses an internal bilinear-forcing result referenced in* `RecognitionScience/Foundation/DAlembert/FullUnconditional.lean`*.*

Full Written Proof

Proof. Define

$$G(t) := F(e^t) \qquad (t \in \mathbb{R}).$$

Then G is C^2, because F is C^2 on $(0,\infty)$ and $t \mapsto e^t$ is smooth. The normalization gives

$$G(0) = F(1) = 0.$$

The reciprocal symmetry of F implies that G is even:

$$G(-t) = F(e^{-t}) = F((e^t)^{-1}) = F(e^t) = G(t).$$

Hence $G'(0) = 0$, and the calibration hypothesis becomes

$$G''(0) = 1.$$

Because G is C^2 and satisfies

$$G(0) = 0, \qquad G'(0) = 0, \qquad G''(0) = 1,$$

its second-order Taylor expansion at 0 is

$$G(t) = \frac{t^2}{2} + o(t^2) \qquad (t \to 0).$$

Therefore $G(t) > 0$ for all sufficiently small nonzero t. Since G is continuous and $G(0) = 0$, it follows that the range of G contains a nontrivial interval $[0, \varepsilon]$. Equivalently, the range of F on $(0,\infty)$ contains a nontrivial interval $[0, \varepsilon]$.

Now use the quadratic symmetric ansatz from the Lean theorem. Because P is symmetric, it can be written in the form

$$P(u,v) = a + b(u+v) + cuv + d(u^2 + v^2)$$

for suitable real constants a, b, c, d.

Set $y = 1$ in the consistency law. Since $F(1) = 0$, we obtain

$$F(x) + F(x) = P(F(x), 0),$$

that is,

$$2F(x) = P(F(x), 0) \qquad (x > 0).$$

Because the range of F contains an interval $[0, \varepsilon]$, this implies

$$P(u, 0) = 2u \qquad \text{for all } u \in [0, \varepsilon].$$

But

$$P(u, 0) = a + bu + du^2.$$

Hence the polynomial identity

$$a + bu + du^2 = 2u$$

holds on an interval. Therefore the two polynomials are identical, so

$$a = 0, \qquad b = 2, \qquad d = 0.$$

Thus P simplifies to

$$P(u,v) = 2u + 2v + cuv.$$

Now evaluate at $(u, v) = (1, 1)$. The normalization $P(1, 1) = 6$ gives

$$6 = P(1, 1) = 2 + 2 + c,$$

so $c = 2$. Consequently,

$$P(u, v) = 2uv + 2u + 2v.$$

Substituting this into the consistency law yields

$$F(xy) + F(x/y) = 2F(x)F(y) + 2F(x) + 2F(y) \qquad (x, y > 0),$$

which is exactly the Recognition Composition Law.

At this point the standard uniqueness theorem applies. The Lean theorem `washburn_zlatanovic_uniqueness_aczel` in `RecognitionScience/Cost/FunctionalEquation.lean` proves that any continuous reciprocal cost satisfying the Recognition Composition Law, normalization, and calibration must equal the canonical cost

$$J(x) = \frac{1}{2}(x + x^{-1}) - 1.$$

Since F is C^2, it is continuous, so we conclude

$$F(x) = \frac{1}{2}(x + x^{-1}) - 1 \qquad (x > 0).$$

It remains to show that the same formula for P holds on the entire first quadrant $[0, \infty)^2$. Fix $u, v \geq 0$, and define

$$x_u := (u + 1) + \sqrt{(u + 1)^2 - 1}, \qquad y_v := (v + 1) + \sqrt{(v + 1)^2 - 1}.$$

These numbers are strictly positive. A direct calculation shows that

$$J(x_u) = u, \qquad J(y_v) = v,$$

because solving

$$\frac{1}{2}(x + x^{-1}) - 1 = u$$

for $x > 0$ gives exactly $x = x_u$, and similarly for v.

Therefore

$$P(u, v) = P(J(x_u), J(y_v)).$$

Using the consistency law with $F = J$, we obtain

$$P(J(x_u), J(y_v)) = J(x_u y_v) + J(x_u/y_v).$$

Now apply the Recognition Composition Law for J:

$$J(x_u y_v) + J(x_u/y_v) = 2J(x_u)J(y_v) + 2J(x_u) + 2J(y_v).$$

Since $J(x_u) = u$ and $J(y_v) = v$, this becomes

$$P(u, v) = 2uv + 2u + 2v.$$

Thus both the scalar law and the combiner are uniquely determined. □

Summary

The supporting Lean chain is:

1. the combiner on the range of F is forced to the RCL form by `consistency_forces_RCL_form_is_theorem`,
2. the scalar law is forced to the canonical cost by `dAlembert_cosh_solution_aczel` and `washburn_zlatanovic_uniqueness_aczel`,
3. the combiner is then forced on all of $[0, \infty)^2$ by `J_surjective_nonneg` and `P_determined_nonneg`.